文件： DeepSeek-V4.1-Flash 模型卡（HuggingFace）
来源： https://huggingface.co/deepseek-ai/DeepSeek-V4.1-Flash
下载/转换时间： 2026-10-21

HuggingFace 官方模型卡正文转换，含 Base/Instruct 评测表、评测设置和复现说明。注意：该模型卡对应 V4.1-Flash（260910）版本，1017 版本分数见官方更新日志。

DeepSeek-V4.1-Flash: Pushing the Limits of KV Cache Compression

DeepSeek-V4.1

Homepage

Technical Report

Introduction

We introduce **DeepSeek-V4.1-Flash** , a multimodal Mixture-of-Experts (MoE) model with 552B backbone parameters and support for contexts of up to one million tokens. The model natively processes images and text, and generates text autoregressively.

Architecture. DeepSeek-V4.1-Flash adopts a **Causal Encoder-Decoder (CED)** architecture: a 40-layer Transformer organized as a 20-layer causal encoder followed by a 20-layer decoder. With CED, the decoder's global KV cache is projected from the final encoder hidden states rather than derived from each decoder layer's own hidden states. This allows the model to activate only **8B parameters per token during prefill** and **16B during decode** , substantially improving cost efficiency for input-heavy agentic workloads. **SWA Bounded Replay** reconstructs missing SWA KV states by replaying only the most recent n_win tokens, avoiding the need to persist SWA KV to SSD and reducing the persistent KV cache footprint to roughly **1/8** of that of DeepSeek-V4-Flash.

Compressed Sparse Attention 2 (CSA2). DeepSeek-V4.1-Flash uses CSA2, which assigns each attention layer one of three static modes — **Full** , **Reindex** , or **Reuse** — to share main KV and indexer K across layers and reuse Top-K sparse-attention indices. In the decoder, a **Hierarchical Sparse Indexer** further restricts later indexing layers to a candidate pool constructed by the first Full Mode layer, bounding deeper indexer cost independently of context length. Combined with **FP4 main KV caching** (E2M1 format, one E4M3 scale per 16 channels), these designs reduce the global KV cache footprint to **890 bytes per token** — roughly **1/4** of DeepSeek-V4-Flash.

Additional architectural components include Single-Pass mHC (revised residual-stream mixing with an efficient Mega-mHC kernel), Engram conditional memory (196B parameters, sparsely accessed via token-based lookup), and DSpark speculative decoding (semi-autoregressive draft generation with confidence-scheduled verification). The model uses 1 shared expert and 384 routed experts per MoE layer, activating 6 routed experts per token.

Multimodal architecture. A vision encoder (DeepSeek-ViT, trained from scratch with 2D-RoPE and 3×3 pixel-unshuffle downsampling) and a two-layer MLP projector convert images into visual embeddings, processed jointly with text embeddings from the start of language-model pre-training.

Pre-training. DeepSeek-V4.1-Flash is trained from scratch on a multimodal corpus comprising **45T tokens** , with sparse attention trained at a sequence length of 64K and context extended to 1M tokens at 34T tokens.

Post-training. The post-training recipe follows the standard SFT → RL → on-policy distillation (OPD) paradigm without algorithmic modifications. All substantive changes lie instead in the data pipeline: large-scale automated synthesis of agent tasks and environments with progressive scaling of data, tasks, and rollouts. The model supports a **continuously controllable reasoning effort** setting (integer 1–100) that trades inference cost for accuracy.

Figure 1. (a) Performance of DeepSeek-V4.1-Flash and counterparts on agentic benchmarks. (b) Global KV cache size per token (bytes) across generations of DeepSeek models. DeepSeek-V4.1-Flash achieves approximately 4-fold and 437-fold reductions relative to DeepSeek-V4-Flash and DeepSeek-V1, respectively.

Evaluation Results

Base Model

All base models are evaluated in our internal framework under the same evaluation settings. Scores within 0.3 of each other are considered equivalent.

Benchmark (Metric)	# Shots	DeepSeek-V4-Flash-Base	DeepSeek-V4-Pro-Base	DeepSeek-V4.1-Flash-Base
Architecture	—	MoE	MoE	MoE
# Backbone Params	—	284B	1.6T	552B
# Activated Params	—	13B	49B	8B / 16B
World Knowledge				
AGIEval (EM)	3–5-shot	83.9	**84.4**	83.4
MMLU-Pro (EM)	5-shot	68.3	73.5	**74.1**
C-Eval (EM)	5-shot	92.1	**93.1**	92.1
MultiLoKo (LLM-Judge)	5-shot	42.6	**50.9**	45.5
SimpleQA-Verified (EM)	25-shot	30.1	**55.2**	42.3
SuperGPQA (EM)	5-shot	46.5	**53.9**	53.1
Language & Reasoning				
BBH (EM)	3-shot	86.9	**87.5**	86.1
BBEH (EM)	1-shot	25.4	**29.8**	27.2
DROP (F1)	1-shot	**88.6**	**88.7**	87.9
HellaSwag (EM)	0-shot	85.7	**88.0**	87.2
Code & Math				
BigCodeBench (Pass@1)	3-shot	56.8	59.2	**60.6**
HumanEval (Pass@1)	0-shot	69.5	76.8	**79.4**
GSM8K (EM)	8-shot	90.8	92.6	**93.0**
MATH (EM)	4-shot	57.4	**64.5**	61.1
MGSM (EM)	8-shot	**85.7**	84.4	80.2
Long Context				
LongBench-V2 (EM)	1-shot	44.7	**51.5**	45.2
Multimodal				
MMMU-Pro (EM)	4-shot	—	—	56.5
CVBench (EM)	4-shot	—	—	77.9
DocVQA (LLM-Judge)	4-shot	—	—	95.6
RefCOCO-avg (Acc@0.5)	0-shot	—	—	86.0

Instruct Model

DeepSeek-V4.1-Flash supports a continuously controllable reasoning effort from 1 to 100. All instruct results below use the maximum effort setting (`reasoning_effort=100`). Evaluations use `temperature=1.0, top_p=0.95` .

For code agent benchmarks (Terminal-Bench 2.1/3.0/4.0, DeepSWE v1.1, NL2Repo-Bench, ProgramBench), the model is evaluated with the Minimal mode of DeepSeek Harness and a 1M-token context window. To align with official setup requirements, the mini-SWE harness is used for DeepSWE v1.1, and the Claude Code harness for SEC-Bench Pro. Visual agent benchmarks (Chartography, BabyVision, ZeroBench) use the Claude Code harness with a 512k-token context window. Agent's Last Exam and AutomationBench use their official scaffolds. All agentic evaluations use `temperature=1.0, top_p=0.95`.

Comparison with frontier models (Max reasoning effort)

Benchmark (Metric)	Opus-5.0	GPT-5.6 Sol	K3	GLM-5.3	DS-V4-Pro	DS-V4-Flash	DS-V4.1-Flash
Reasoning							
GPQA Diamond (Pass@1)	93.4	**94.1**	92.9	88.1	92.4	89.9	90.9
HLE (Pass@1)	**56.3**	44.5	43.5	42.0 †	42.7 †	37.8 †	36.8 (39.1 †)
Codeforces (Rating)	—	—	—	—	3348	3289	**3471**
MathArena Apex (Pass@1)	—	—	**65.6**	—	65.3	58.6	**65.6**
Agentic							
Terminal-Bench 2.1 (Pass@1)	89.1	88.8	88.3	88.2	87.9	82.7	**90.6**
Terminal-Bench 3.0 (Pass@1)	**43.3**	34.4	17.7	28.3	11.8	7.6	30.0
Terminal-Bench 4.0 (Pass@1)	**51.8**	39.9	12.6	37.9	12.4	7.0	31.2
DeepSWE v1.1 (Resolved)	74.0	73.0	67.5	66.9	62.7	54.4	**74.2**
ProgramBench (Almost@1)	**37.0**	23.0	17.5	19.0	15.5	—	20.3
NL2Repo-Bench (Score)	**75.3**	56.8	58.0	58.0	61.5	54.2	64.0
CyberGym (Pass@1)	—	84.5	80.0	84.5	83.3	76.7	**88.1**
SEC-Bench Pro (Pass@1)	—	**74.3**	—	—	56.4	30.9	62.8
ExploitGym (Pass@1)	22.1	**33.7**	—	15.0	5.4	1.8	15.3
HLE w/ tools (Pass@1)	63.6	—	59.8	62.5	60.0	51.5	**63.9**
AutomationBench (Pass@1)	50.3	45.8	46.7	48.8	43.2	37.7	**54.8**
Agent's Last Exam (Pass@1)	28.6	26.7	27.6	28.5	25.7	25.2	**31.8**
Chartography w/ tools (Pass@1)	**84.0**	79.9	68.1	—	—	—	78.9
BabyVision w/ tools (Pass@1)	**94.1**	88.9	85.7	—	—	—	89.6
ZeroBench-main w/ tools (Pass@5)	52.0	**53.0**	41.0	—	—	—	49.0

† Text-only subset of HLE.

Performance across agent scaffolds (DeepSWE v1.1 and Terminal-Bench 2.1, Max reasoning effort)

All scaffolds use N=8 samples per task on DeepSWE v1.1 and N=3 on Terminal-Bench 2.1, with Linux containers, `temperature=1.0`, `top_p=0.95`, a 1M-token context limit, and max_steps=500 per agent. Terminal-Bench 2.1 is evaluated without network access.

Benchmark (Metric)	Claude Code	Codex	OpenCode	Pi	mini-SWE	DSH Minimal	DSH Standard	DSH PTC
DeepSWE v1.1 (Resolved)	69.8	65.6	65.5	66.2	74.2	72.6	70.5	67.6
Terminal-Bench 2.1 (Pass@1)	88.0	84.1	85.0	86.1	90.3	90.6	85.8	85.8

Prompt Encoding

This release does not include a Jinja-format chat template. The `encoding` folder contains a self-contained Python reference implementation (`encoding.py`) with test cases for multi-turn conversations, tool calling, thinking mode, numeric reasoning effort, mid-conversation system messages, and interleaved image content.

For production use, we additionally release deepseek-recipe, a set of Rust libraries with Python bindings that provides the same prompt format as a maintained, protocol-aware toolkit. It converts Messages, Chat Completions, and Responses API requests into the Conversation format, encodes them into DeepSeek V4 and V4.1 prompts or token IDs, and parses model output back into complete or streamed responses — covering thinking, tool calls, images, and generation settings. Model inference, tool execution, and HTTP transport are left to the caller.

Minimal Inference

Please refer to the `inference` folder for instructions on weight conversion and running inference locally.

Recommended sampling parameters:

Parameter	Value
`temperature`	1.0
`top_p`	0.95 or 1.0
`context_window`	1M tokens
`max_tokens`	⩾ 256K

Reproducing DeepSWE Benchmark Results

The `evaluation` folder contains step-by-step instructions for reproducing the DeepSWE v1.1 benchmark results, covering both the `dsh-minimal` agent and the official `mini-swe-agent`. The patch required to integrate `dsh-minimal` with Pier is also included there.

License

This repository and the model weights are licensed under the MIT License.

Citation

```
@misc{deepseekai2026deepseekv41flash,
 title={DeepSeek-V4.1-Flash: Pushing the Limits of KV Cache Compression},
 author={DeepSeek-AI},
 year={2026},
}
```

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763B params

Tensor type

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F32

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Model tree for deepseek-ai/DeepSeek-V4.1-Flash

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20 models

Quantizations

101 models